CAP Theorem: Use It to Choose an Open Source Database

This is a practical guide to choosing the right open source database. This choice depends on a clear understanding of the CAP theorem and the unique characteristics of various open source databases.

Data surrounds us in every aspect of our lives, thanks to the pervasive role of technology. From the devices we hold in our hands to enterprise databases, data storage has become essential. The days of relying on a few well-known companies for data storage are long gone, and we now have a plethora of choices when it comes to selecting databases for different use cases. However, with so many options, especially in the open source realm, it can be overwhelming to evaluate the right database for your needs. This article aims to guide you in making that decision.

Deploying modern applications in the cloud, with a distributed architecture across multiple cloud environments, has become a common practice for organisations seeking high availability, performance, and resiliency. While this approach is beneficial for meeting non-functional requirements, a deeper look at database deployment would have to be considered in a distributed deployment. Factors such as performance, characteristics and properties of the database are important when selecting a product for specific use cases. Over the past two decades, experts in database technology have adopted proven methods to evaluate databases. The CAP theorem is one of the most popular methods for screening databases, especially for open source databases, where so many options are available. Let's explore how the application of the CAP theorem in data and transaction models compares for some of the most popular open source databases.

Distributed databases and the CAP theorem

In modern application design, distributed architecture is widely employed, where databases are deployed across multiple interconnected locations to achieve higher resilience and performance. While this approach is crucial for various reasons, it is equally important to evaluate and select the right database that works for specific use cases. When evaluating a database, we need to consider what we expect from the database out-of-the-box and what we can address through design in terms of consistency, availability and partition tolerance. Let's take a quick look at what each of these characteristics means and how we can apply them while evaluating a database in a distributed system.